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Review Article

A Deep Learning Approach for Type-2 Diabetes Risk Prediction – A Review

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ABSTRACT

Searching the vicinity of the fuzzy boundaries, type 2 diabetes is one of the most widespread chronic diseases that demand early identification for its effective treatment and prevention from severe complications. The advent of machine learning (ML) and deep learning (DL) as predictive healthcare techniques has put forward the opportunity for superior accuracy than traditional statistical approaches. This survey paper endeavors to include various algorithms used in ML and DL for prediction in Type 2 diabetes, their methods of operation, and broad performance analysis with challenges encountered therein. A survey of some of the ML models such as K-Nearest Neighbors (KNN), Random Forest, Support Vector Machines (SVM), Decision Trees, Logistic Regression, and ensemble methods along with the DL techniques like Recurrent Neural Networks (RNNs), Convolution Neural Networks (CNNs), Artificial Neural Networks (ANNs), Long Short-Term Memory (LSTM) is presented in this paper. This study, further, discusses the importance of feature extraction, pre-processing, hyper-parameter tuning, and regularization techniques that act as add-ons to bolster one's performance. We also review benchmark datasets the Pima Indians Dataset and the datasets from clinical studies applicable to diabetes prediction. The paper concludes with observations on current challenges, future research directions, and the incorporation of explainable AI (XAI) techniques for the enhancement of interpretability and clinical applicability of the model.

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INTRODUCTION

A common chronic illness, type-2 diabetes mellitus (T2DM) has a number of serious side effects, such as neuropathic disorders and renal and cardiovascular insufficiency. According to the World Health Organization, the prevalence of diabetes has nearly quadrupled worldwide in the past few decades, clearly indicating an urgent requirement for effective predictive methods to tackle and mitigate its severity. Proactive risk assessment for Type 2 diabetes is useful to health care professionals while recognizing individuals at elevated risk, thus allowing for timely intervention and management strategies targeted at preventing or delaying the onset of the disease. Traditional statistical approaches have long been used in predicting diabetes risk, but they often do not maintain well in representing the complicated and nonlinear relationships among genetic, demographic, clinical, and lifestyle factors that underlie T2DM risk. In contemporary times, deep learning has surfaced as a powerful methodology for intricate predictive functions within the healthcare domain, utilizing extensive datasets to uncover patterns that might remain concealed from conventional modeling techniques. Nevertheless, a significant challenge associated with deep learning in medical applications pertains to the interpretability of both the models and their resultant predictions. Health care data typically are heterogeneous in nature, containing structured data like lab results and demographic information, as well as unstructured data, which include clinical notes. Researchers have studied semantic approaches that will allow the representation of meaning and contextual information within the data, thereby enabling the improvement of deep learning models through the provision of interpretative layers that will help understand the interrelation between various variables. This paper follows recent research efforts in the context of deep learning methods used for prediction tasks associated with Type-2 diabetes risk, focusing especially on methods that incorporate semantic methods. Semantic methods-including word embeddings, ontology-based methods, and semantic net methods-describe medical concepts and interrelations more predictively and interpretable ways. This research approach tries to combine these approaches with deep learning frameworks to more effectively capture the complex and context-dependent nature of factors determining diabetes risk. The target of this survey is to provide a structured overview of deep learning models that are enhanced using semantic methods in the prediction of T2DM, outlining common trends and approaches, and discussion of the pros and cons of various models. We discuss various architectures of neural networks, including CNNs, RNNs, and hybrids that exploit semantic knowledge for better performance as well as better explainability. This review discusses the extant issues in the area as well, including data quality, interpretability, and the integration of different sources of data, and gives hints on possible future developments. With this view, through this paper, we attempt to contribute to the understanding and promotion of the use of semantic-rich deep learning methods for type 2 diabetes risk prediction by investigators and clinicians.

1. Related Work

Ram D. Joshi and Chandra Dhakal suggested a machine learning model that uses decision trees and logistic regression to forecast when Pima Indian women will develop type 2 diabetes. The classification tree model focuses extensively on glucose, BMI, and age, other models put much emphasis on pregnancy and diabetes pedigree function as significant predictors [1]. Using a multi-layer neural network with no-prop algorithm, J. Jeba Sonia, Prassanna Jayachandran,

Abdul Quadir Md, Senthilkumar Mohan, Arun Kumar Sivaraman, and Kong Fah Tee have suggested a novel system of classification of the three forms of diabetes. The neural network was trained and attributes were chosen differently for each kind of diabetes in the two-phase algorithm work-in-training and testing phases. Normal and type 1 diabetes were used to train the neural network's first layer, followed by normal and type 2 diabetes, healthy and pregnant diabetes, and finally normal and type 2 diabetes [2]. Neha Prerna Tiggaa, Shruti Garga have built machine learning models for the classification of the type 2 diabetes using six algorithms. They tested various dataset results and a set collected online and offline, using 18 diabetes-related questions, and on the PIMA database. The most significant predictors found were family history of diabetes, regular medication, age, physical activity and gestational diabetes [3]. Using characteristics from the current year (Y), Henock M. Deberneh and Intaek Kim created a machine learning (ML) model to forecast the occurrence of T2D in the year that follows (Y + 1). The researchers used recursive feature reduction techniques, chi-squared tests, and ANOVA testing to identify important predictive features in electronic health records spanning 2013 to 2018. The model for classifying a person as non-diabetic, prediabetic, or diabetic was constructed using the majority of machine learning methods. The study stresses how crucial it is to include a variety of health metrics and how these models might support clinical judgment and early T2D intervention [4]. A model developed by Zhao, Yuedong, et al. performed well in forecasting the risk of DR in patients with type 2 diabetes mellitus at every time point. This study demonstrated how the XGBoost model may help physicians identify high-risk patients and make decisions about the management of type 2 diabetes [5]. Chaitanya Krishna, Suryadevara developed a system that, by combining the findings of several machine learning techniques, can accurately determine whether a patient has diabetes. Here the practitioner built the models using machine learning algorithms like decision tree, random forest, Support Vector Machine, and Logistic regression [6]. Islam Safial Ayon, Md. Milon Islam, suggested a method that uses medical information to diagnose diabetic patients. By employing five-fold and ten-fold cross-validation to train the deep neural network's properties, they suggested a method for diagnosing diabetes [7]. Dr. Vaidehi and Aishwarya Mujumdar have compared the accuracy of machine learning algorithms using two distinct datasets. In order to better classify diabetes, they have suggested a diabetes prediction model that incorporates a few external elements that contribute to the disease in addition to standard parameters like age, insulin, BMI, glucose, and so on. When compared to existing datasets, classification accuracy is improved with fresh datasets [8]. Huaping Zhou, Raushan Myrzashova, and Rui Zheng presented a technique that can be used to forecast the future incidence of diabetes and identify the sort of condition the individual would have. They discovered that the therapy approaches for type 1 and type 2 diabetes varied greatly, and they guarantee that their suggested approach will assist in giving the patient the appropriate care. Deep neural networks were used in the model-building process [9]. Tasin Isfafuzzaman using a private dataset of female patients in Bangladesh and a variety of machine learning approaches, Tansin Ullah Nabil Sanjida Islam Riasat Khan created an autonomous diabetes prediction system. To reduce the problem of class imbalance, ADASYN (Adaptive Synthetic Sampling) and SMOTE (Synthetic Minority Over-sampling Technique) were used. To ascertain which algorithm yields the best prediction results, practitioners employed machine learning classification algorithms, including decision trees, SVM, Random Forest,

Logistic Regression, KNN, and various ensemble approaches [10]. A strong pipeline for preparing datasets pertaining to diabetes was created by Talukder, Md. Alamin, et al. According to their findings, the XGBoost model performed exceptionally well on the Tigga dataset, while the RF model performed well on the Austin Public dataset and the Pima Indian dataset. On the Mendeley dataset, the DT model performed flawlessly [11]. Liu, Lianhua, et al. utilized machine learning techniques to predict complications associated with type 2 diabetes, particularly focusing on peripheral vascular disease (PVD) also best features subset was chosen for the prediction model development using RFE with 5-fold cross-validation. Development and validation of predictive models, with the XGBoost model demonstrating superior performance in predicting PVD risk among hospitalized patients [12]. Sarkar, Prosanjeet Jyotirmay, et al. discusses a model for predicting diabetes mellitus (DM) using machine learning techniques applied to the NHANES dataset. It evaluates five algorithms, including XGBoost-histogram, which outperformed others in accuracy and AUC-ROC, achieving a cross-validation accuracy of 86.4%. The study emphasizes the significance of lifestyle factors in diabetes prediction and proposes a user-friendly model for real-time predictions that can be integrated into mobile applications, aiming to enhance early detection and reduce healthcare costs associated with diabetes management [13].

2. DIABETES AND ITS TYPES

Diabetes is a long-term illness that alters how your body uses food as fuel. Normally, food is converted by the body into glucose, or sugar, which is subsequently released into the bloodstream. The hormone insulin, which is produced by the pancreas, facilitates the entry of glucose into the body's cells for energy. There are so many types of diabetes that can cause many chronic diseases, some of the types are:

 The immune system of the body targets and kills the beta cells in the pancreas that produce insulin in type 1 diabetes, an autoimmune disease. It can happen at any age, but it usually manifests in kids, teens, or young adults. symptoms that appear suddenly, including impaired vision, acute fatigue, weight loss, excessive thirst, and frequent urination.

 When the body grows resistant to insulin or the pancreas is unable to produce enough insulin to keep blood glucose levels within normal ranges, type 2 diabetes develops. Lifestyle factors, such as obesity, lack of physical activity, and poor diet, are major contributors. Although type 2 diabetes typically strikes adults over 40, rising obesity rates are making it more prevalent in younger people, including youngsters. It’s symptoms often develop gradually and may include fatigue, increased thirst, frequent urination, blurry vision, slow-healing sores, and frequent infections.

 High blood sugar levels are the result of gestational diabetes, which happens during pregnancy when the body is unable to create enough insulin to satisfy the increasing needs. It is believed to be brought on by hormonal changes that occur during pregnancy and reduce the body's sensitivity to insulin. It commonly appears between weeks 24 and 28 of pregnancy and goes away after giving delivery. Often, there are no noticeable symptoms, but some women may experience excessive thirst, frequent urination, or fatigue.

 Blood glucose levels that are higher than normal but not high enough to be classified as Type 2 diabetes are known as prediabetes. It frequently precedes Type 2 diabetes, which means that if

treatment is not received, it may develop into diabetes. Prediabetes can develop over many years and often has no obvious symptoms. Like Type 2 diabetes, prediabetes often has no symptoms, but a blood test will reveal elevated glucose levels.

3. Deep Learning Methods

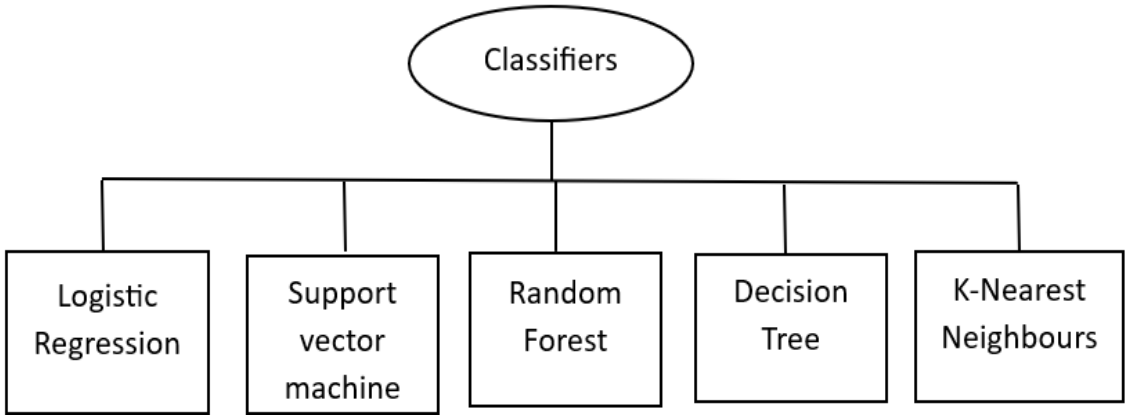
According to medical professionals and recent studies, the likelihood of recovery is higher if the illness is identified early. As technology continues to progress, machine learning and deep learning methods have emerged as highly valuable tools for illness investigation and early prediction. Among these methods, the Convolutional Neural Network (CNN), Random Forest (RF), and Support Vector Machine (SVM) are employed in this study to forecast diabetes. Using machine learning and deep learning techniques to predict diabetes has been the topic of several recent studies. Deep learning techniques for diabetes prediction use advanced neural networks that analyse medical data to find out the risk patterns of diabetes. The key approaches we use can be feedforward neural networks (FNNs) for structured data, RNN and LSTMs for the time-series glucose trend, and CNNs for the medical imaging like retinal scan. Techniques like Transformers and Graph Neural Networks (GNNs) improve text and relational data analysis. Autoencoders and GANs manage feature extraction and data augmentation, while Attention Mechanisms and Explainable AI improve interpretability. Multimodal learning integrates diverse data sources, and Reinforcement Learning aids personalized treatment. These methods enable early diagnosis, glucose prediction, and treatment planning, overcoming challenges like data imbalance, privacy, and model interpretability.

4.1 Deep Neural Networks approach for prediction

In [9], Rui Zheng, Raushan Myrzashova*, and Huaping Zhou employed a systematic deep learning framework. For the classification, a deep neural network was employed. The method feature engineering, in which a new feature vector is created by concatenating, processing, and immediately feeding the feature vector into input nodes. Different activation functions are used across layers, and a SoftMax classifier at the output layer determines classification confidence. The model undergoes training and optimization, starting with random weight initialization, followed by sequential fine-tuning, loss function minimization, and accuracy maximization [9].

4.2 Classification approaches

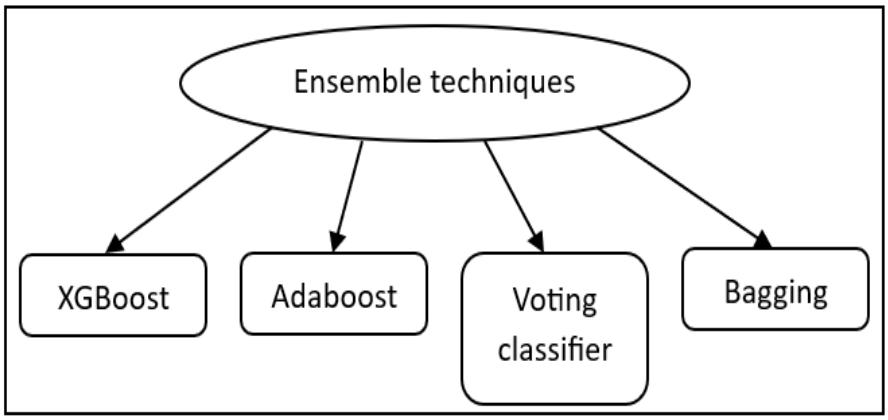
There are various classifiers exists in machine learning used for the diabetes classification.



The approach for automatic diabetes prediction in the given text involves employing different ensemble and machine learning methods, optimized using Grid Search CV to find the best hyperparameters and prevent overfitting [10]. Several classifiers were implemented, including Decision Trees, which use Gini impurity or entropy for splitting nodes, optimized with max depth = 2 and minimum samples per leaf = 50. The K-Nearest Neighbors (KNN) classifier determines classification using K = 5 neighbors based on distance measures. Although KNN is a sluggish predictor in comparison to other methods, it can be utilized to address classification and regression problems [6]. Random Forest, an ensemble model averaging multiple decision trees, was applied with 400 estimators and minimum samples per leaf = 5. Support Vector Machines (SVM) were tested with various kernels, with the linear kernel (C = 10, gamma = 1) performing best. Logistic Regression, a binary classification model, was optimized with 150 iterations for convergence [10].

4.3 Ensemble Techniques

With 50 estimators and a learning rate set at 0.10, the AdaBoost ensemble technique adjusted the weights assigned to misclassified instances. With max depth = 4 and binary logistic as the objective function, the gradient-boosted decision-tree model XGBoost was run [10,12]. Voting Classifier, an ensemble method of models working in soft voting regarded as one of the better classifying models, was used [10]. Bagging Classifier made several predictions by aggregating outputs from randomly-specified base classifiers. All the techniques have then combined all are for improved classification accuracies on Diabetes Prediction performance through hyperparameter tuning, ensemble learning, and optimization mechanisms.

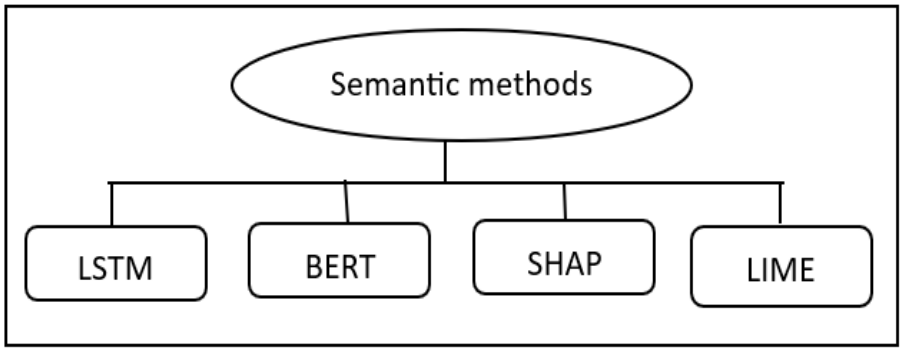


The bagging techniques showed best accuracy performance in the diabetes classification [10].

4. Semantic Methods

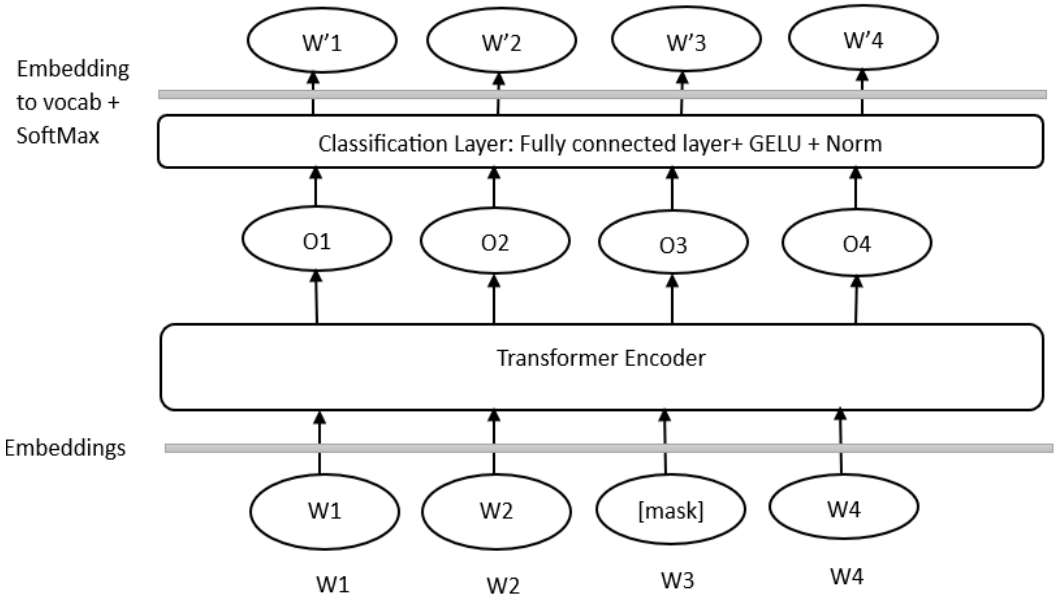
Semantic methods in deep learning for diabetes prediction focus on understanding and interpreting the complex, often unstructured medical data to identify patterns or insights associated with diabetes risk. These methods aim to leverage semantic representations of data for accurate prediction, classification, and insight generation. Semantic approaches to deep learning in diabetes prediction include insights about extracting relevant findings from complex medical data sets. Techniques used include embeddings (such as BERT, Bio BERT for the analysis of clinical text), temporal models (like LSTMs, Transformers), which extract time-series patterns in the data, and knowledge graphs to connect structured and

unstructured data sources. They apply attention mechanisms to point to critical predictors, contextualized representations for health records, and combining multimodal data, including text from clinical notes and images. Such Explainable AI techniques as SHAP and LIME ensure model interpretability, hence facilitating early detection, tailored treatment, and lifestyle recommendations. In summary, semantic approaches enhance precision and decision-making regarding diabetes prediction.



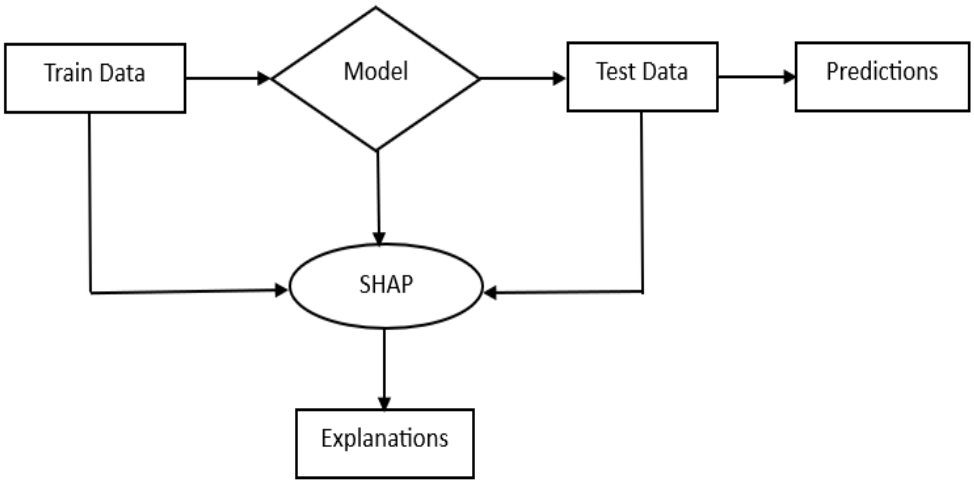
LSTM: Long Short-Term Memory networks is a type of Recurrent Neural Network which is highly effective for sequence-based data modelling due to their ability to learn long-term dependencies. For diabetes prediction, LSTMs can analyse time-series data, such as glucose levels, insulin levels, and other health metrics, to control the course of diabetes or forecast the risk of developing it.

BERT: Bidirectional Encoder Representations from Transformers leverages its powerful natural language processing capabilities. BERT can analyse textual or sequential health records, such as clinical notes, patient histories, or survey responses, to predict diabetes or its progression.

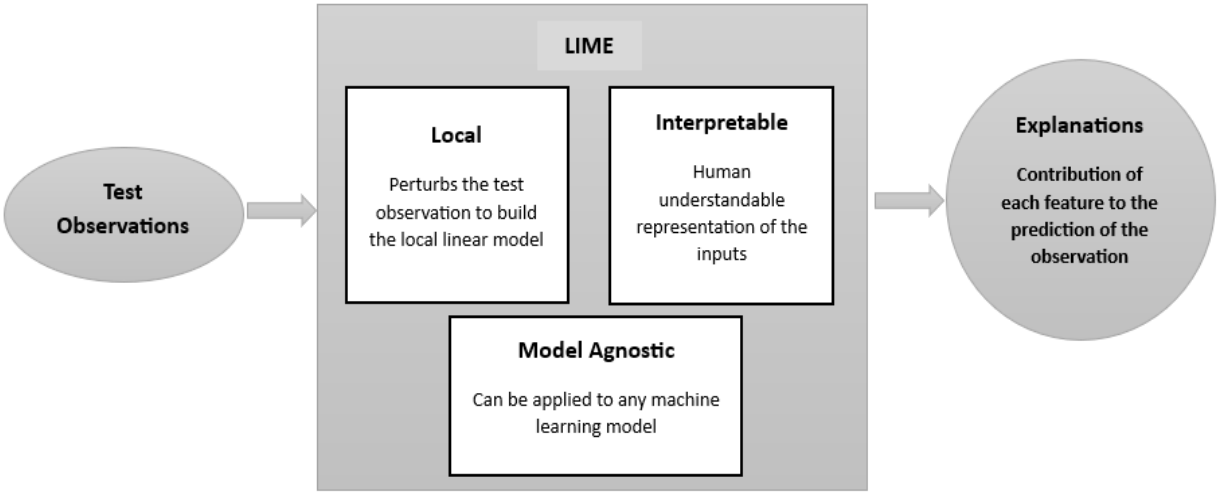


SHAP: Shapley Additive explanations method is a powerful tool for interpreting machine learning models. For diabetes prediction, SHAP provides insight into how individual features (e.g., glucose,

BMI, age) contribute to a model's predictions, improving interpretability and trustworthiness in medical AI applications. The contributing factors of a person acquiring PVD can also be examined using the SHAP approach [12]. Using the SHAP and LIME frameworks, explainable AI techniques are used to comprehend how the model makes the prediction. Using the SHAP library and explainable AI, the XGBoost model with ADASYN feature importance was developed [10,12].



LIME: The process of using interpretable models to locally approximate machine learning model predictions is known as "Local Interpretable Model-Agnostic Explanations." For diabetes prediction, LIME can provide valuable insights into which features (e.g., glucose, BMI, age) contributed most to a specific prediction.



The LIME explainable AI method's implementation of the XGBoost model. The study claims that the model has an 80% confidence level in accurately predicting diabetes in the individual. [10].

CONCLUSION

Integration of semantic methods with deep learning models seems promising for further improvements in the prediction of risks for Type-2 diabetes. Traditional ML methods like Logistic

Regression, Decision Trees, Random Forest, and SVM continue to deliver reliable predictions with interpretable results [1,6,10]. Advanced ensemble techniques such as XGBoost and hybrid models exhibit superior performance, particularly in handling large and complex datasets [9,11,12,13]. In this study I found that Glucose levels, BMI, age, and pregnancy history consistently emerge as significant predictors of diabetes [8] External factors such as physical activity, family history, lifestyle habits, and gestational diabetes enrich prediction models and enhance accuracy [3]. Recursive Feature Elimination (RFE) and statistical tests like ANOVA and chi-squared are widely used for optimal feature selection [4,12]. Studies utilize publicly available datasets (e.g., PIMA, NHANES) and private datasets to train and validate models, also robust preprocessing techniques, including SMOTE and ADASYN, address challenges such as class imbalance, improving model performance and fairness [10]. The survey presented herein underlines the possibility and potentiality of semantic methods, like word embeddings, ontology-based methods, and contextual networks. Based on a comprehensive evaluation of the methods, the study establishes their strengths, weaknesses, and applicability in handling complex healthcare data. It further identifies challenges such as quality of data, interpretability, and integration, offering a basis for future work. In this survey we have seen that the results of XGBoost model has given the superior performance over the traditional machine learning algorithms [11,12]. Most of the papers has utilized the PIMA Indians Diabetes dataset for their project. The ADASYN approach was utilized with XGBoost gives the best results compared to the SMOTE [10]. This literature review will become a valuable reference for the research community and practitioners as well, and it shall inspire the application of rich semantics deep learning techniques for prevention healthcare and Type-2 diabetes management.

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